

DHARANI KUMAR

EMBEDDED SOFTWARE DEVELOPER

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CAREER OBJECTIVE

To secure a challenging role in software development where I can leverage my technical expertise and problem-solving skills to contribute to organizational success while advancing my professional growth.

PROFESSIONAL SUMMARY

- 3+years Experienced in Linux-based embedded software development using C, Embedded C, C++11.
 - Practical experience applying C++11 OOP and advanced features (templates, smart pointers, lambdas, move semantics, STL) in embedded system development.
 - Practical experience developing embedded GUI applications using Qt Widgets and Qt QML frameworks.
 - Well-versed in Linux Internal concepts like IPC, socket programming, multithreading, etc.
 - Applied SQLite integration techniques to ensure smooth interaction between backend modules and GUI components
 - Experienced in developing and optimizing embedded applications on ARM Cortex-A processors within Linux environments.
 - Experienced in developing low-level drivers for communication protocols such as UART, I2C, SPI and TCP/IP.
 - Strong proficiency in Embedded C programming for hardware interfacing, including GPIO, ADC/DAC, LCD, and other peripheral devices.
 - Gained a solid understanding of the Software Development Life Cycle (SDLC) and Agile methodology.
 - Strong analytical and problem-solving skills, with the ability to troubleshoot complex embedded systems and deliver efficient solutions under tight deadlines.
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PROFESSIONAL TOOLS

- **Languages** : C, C++, Embedded C, Linux internals, QML.
- **Microprocessors** : ARM Cortex-A
- **Communication Protocols** : UART, SPI, I2C, TCP/IP
- **Tools Known** : Keil IDE, QT Creator, SQLite
- **Operating Systems** : Linux&Windows

WORK EXPERIENCE

Embedded Software Developer – Jan 2023 – Till date

Creative Tech Mars Software Solutions Pvt. Ltd.

PROJECTS

1.Smart Pharmaceutical Distribution Controller with Embedded Tracking Capabilities

Description:

Touch sensitivity automated system used to track pharmaceutical substances during distribution. The system monitors material levels in real time and alerts operator for refilling before gets exhausted. It also ensures accurate smooth operation, and secure storage of distribution data for easy tracking and reporting.

Tech Used: Embedded C++,Qt Creator,Ubuntu,ARM Cortex-A series

Role: Developer

Responsibilities:

- Analysed technical requirements to design software modules and ensure compliance with system specifications.
- Developed error-handling and recovery mechanisms to maintain smooth dispensing operations under edge conditions.
- Integrated backend modules with the GUI to ensure seamless operator interaction and accurate status reporting.
- Conducted code reviews and collaborated with peers to maintain coding standards and improve overall software quality.

2.Embedded Processor-Powered Solution for Secure and Scalable Information Module

Description:

It is build to streamlining secure and efficient digital data exchange, with a strong focus on data integrity. It ensures that information remains easily accessible in digital format, offering flexibility and scalability for diverse use cases.

Tech Used: Qt Creator,C++11,Linux,SQLite and TCP/IP

Role: Developer

Responsibilities:

- Collaborated in drafting and conceptualizing product designs for technical and functional planning.
- Designed and refined QML user interfaces to enhance usability and meet requirements.
- Implemented core backend features and integrated software models for synchronization and communication.
- Conducted testing and debugging to validate functionality to ensure reliable performance.
- Coordinated with team members to align expectations and communicated timelines to ensure smooth project delivery.

3.Sensor-Integrated Embedded Environmental Management System

Description:

Developed an embedded solution to monitor temperature, humidity, and soil moisture, and automatically control a water pump. Sensor data was displayed on a graphical LCD for real-time local monitoring, ensuring efficient irrigation management and user-friendly visualization.

Tech Used: ARM Processor,UART,GLCD,I2C,Embedded C.

Role: Developer

- Interfaced environmental sensors with a microcontroller to enable real-time data acquisition and actuator control.
- Integrated UART communication for seamless data transfer between microcontroller and peripheral modules.
- Involved in graphical LCD interfacing with microcontrollers using Embedded C.
- Applied efficient memory management techniques to optimize performance of device.

EDUCATION

B.Tech – Electronics and Communication Engineering

JNTU College of Engineering, Pulivendula – 2023 – CGPA: 7.5

Intermediate

Vani Junior College,Tadipatri – 2019 – CGPA: 10.0

PERSONAL DETAILS

- **Date of Birth** : 02/06/2002
- **Nationality** : Indian
- **Language** : English,Telugu,Hindi
- **Marital Status** : Unmarried
- **Relocation** : Pan India

I hereby declare that all the above information is true and correct to the best of my knowledge.

Date:

Dharani Kumar